New Eagle™
MECHATRONIC CONTROL SOLUTIONS

Rinehart Motion Systems PM100

MotoHawk CAN Library

MotoHawk

Part # - MHAWK_RMS_PM100

Version 2

Created by:

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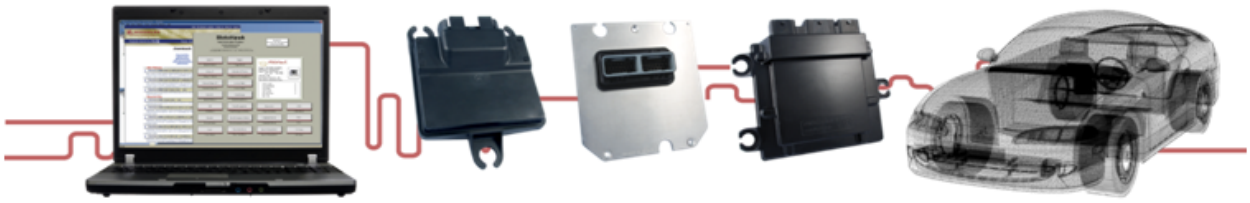

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1.0 CAN Library Background Information

The CAN library is designed as the communication tool between the Rinehart Motion System PM100 Motor Controller (RMC) and the New Eagle Motohawk Supervisory Controller (MSC). Up-to-date system configuration and status of the motors is transmitted via the CAN bus at 250 kbaud rate from the RMC to the MSC on CAN channel1. As part of the supervisory controls, the MSC will receive the broadcast messages and transmit motor commands and parameter commands.

Testing of functionality of the library took part in two phases. Phase 1 included testing the command messages through a CAN Bus Analyzer and MotoTune. The CAN Bus Analyzer verifies that the broadcast messages (0x0A0 – 0x0AC) from the RMC, and the command messages from the MSC, were sent at the correct frequency, data format and content.

Phase 2 included testing the Parameter Messages (0x0C1 and 0x0C2). 0x0C1 is a message sent by the MSC commanding the RMC to read or write data to an address. 0x0C2 is the response message from the RMC detailing whether or not the read or write command was successful. These addresses range from 0-99 and 100-499. Further information regarding the functionality of these messages can be found in the Rinehart Motion System User Manual for the PM100.

1.1 CAN Bus Analyzer and Available Probes, Calibration Tools in MotoTune

CAN King was used to test that the broadcast messages from the RMC are transmitted at the correct frequency. As an example used for testing, Message 0x0A0 was broadcast by a second MotoTron controller that substituted the Rinehart Motion Controller. 0x0A0 contains transistor gate temperature readings, and the message was successfully transmitted every 100ms (Figure 1).

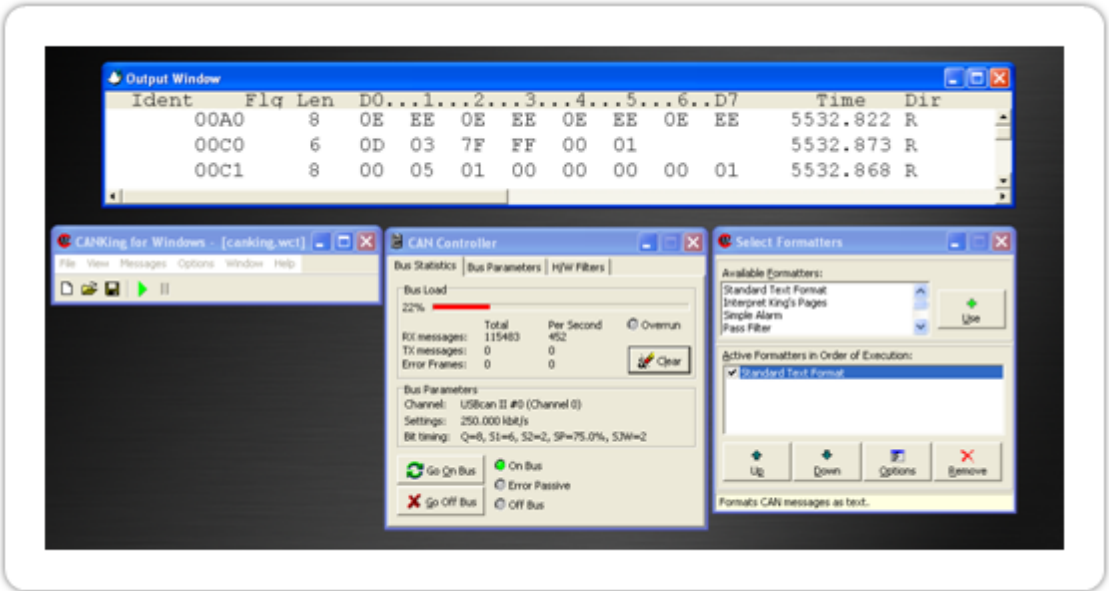

Figure 1: CAN King Traffic Shows Message 0x0A0 On Bus and Transmitting at 10 Hz.

MotoTune played an important role while testing the CAN logic of the supervisory controller. The broadcast messages sent by the motor controller, were set as probes on the MSC display (Figure 2). On the other hand, the command messages transmitted by the MSC are calibration tools in which the data can be varied and set in real time and available for viewing as probes on the RMC display (Figure 3). These two displays are identified in Figure 4 as PCM-1, the Rinehart Motor Controller, and PCM-2 the Motohawk Supervisory Controller.

As an example, temperature readings were transmitted by the RMC to the MSC through message 0x0A0. On the RMC side, calibration tools were used to set broadcast information, and in the case of message 0x0A0 to set temperature values. To verify that the broadcast message was received with the appropriate information by the MSC, the temperature probes for 0x0A0 were placed on the PCM2 display for real-time monitoring. The probe readings of 49.92 degrees C reveal that the information was being broadcast correctly by the RMC and received correctly by the MSC (Figure 4).

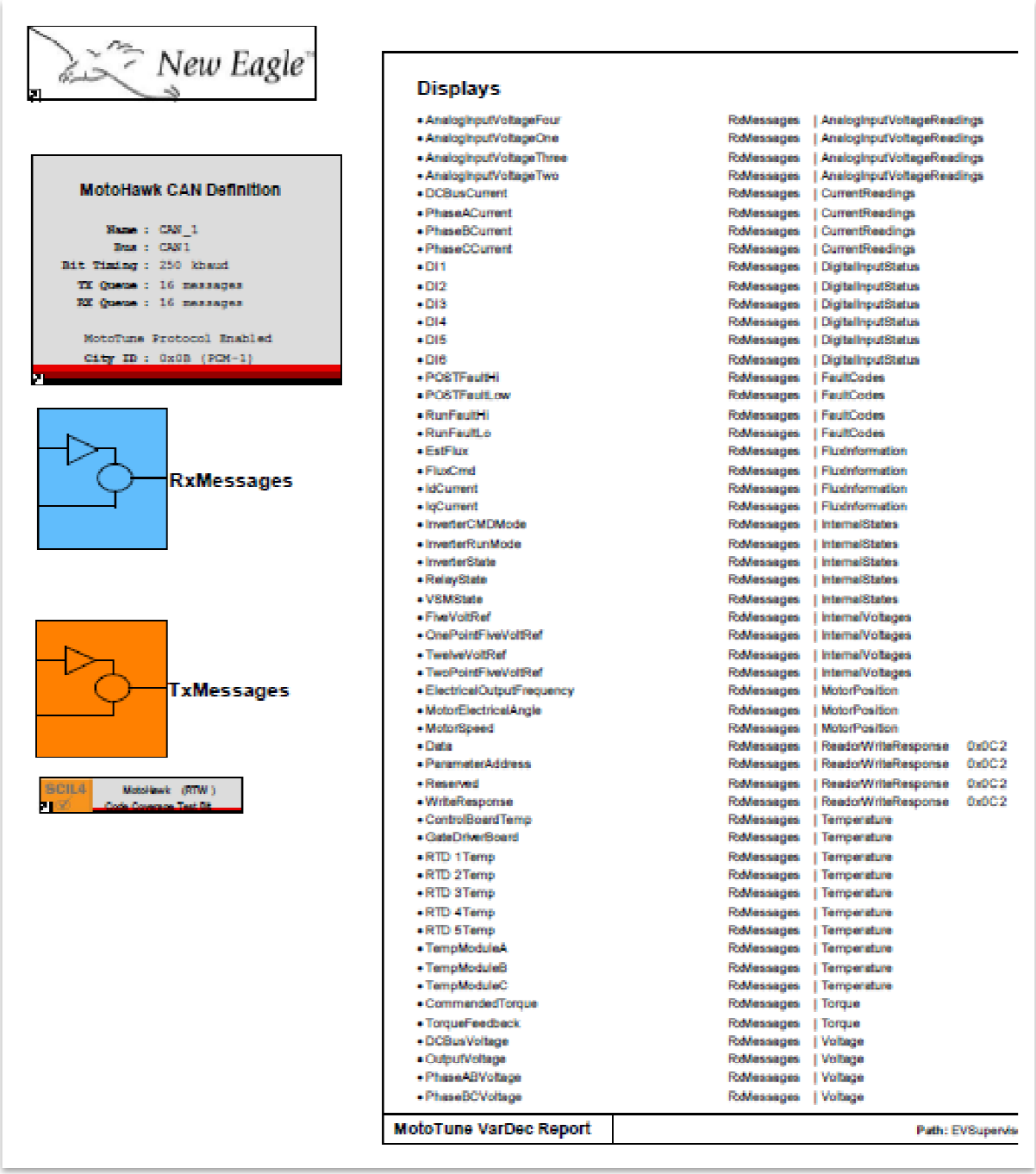

Figure 2: Available Probes in MotoTune for Received Messages from Rinehart Controller.

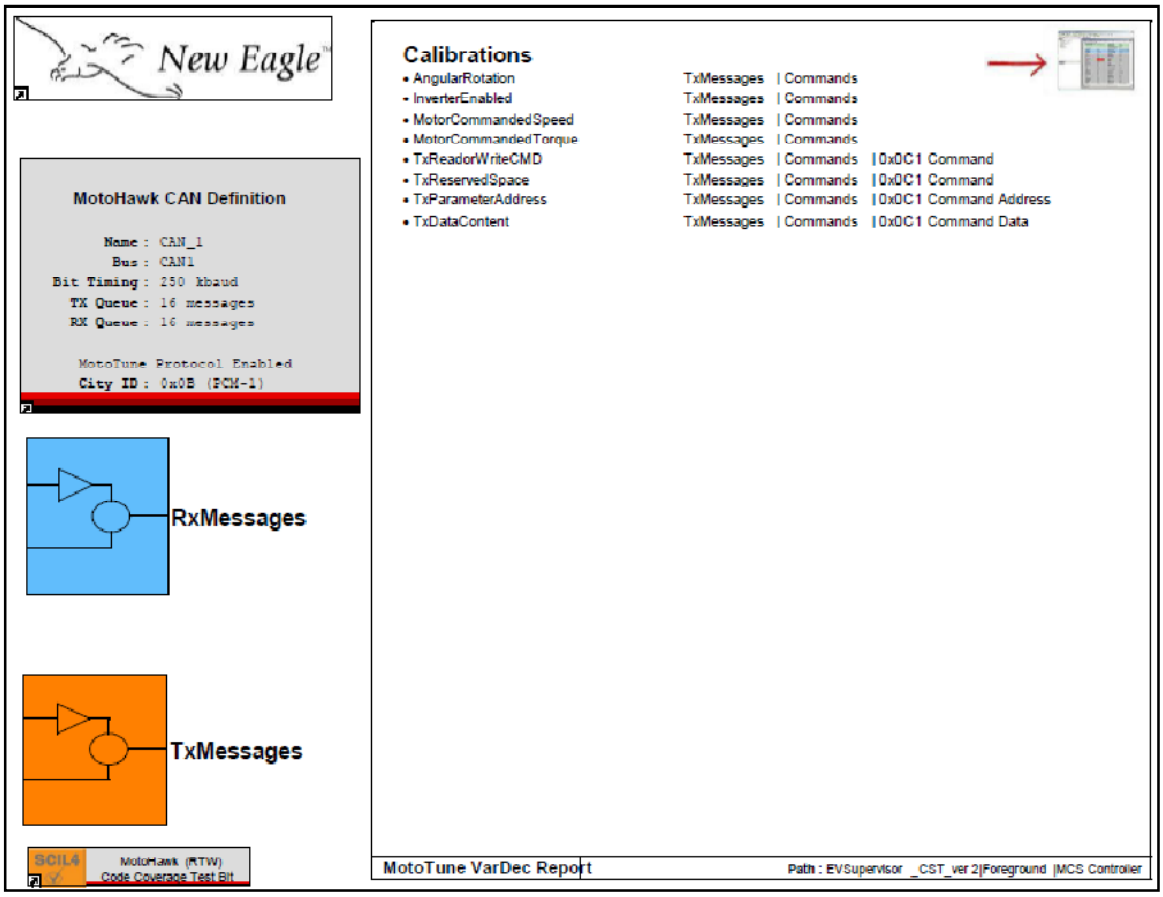

Figure 3: Available Calibration Tools in MotoTune for Transmitted Messages from MSC.

1.2 Parameter Messages 0x0C1 and 0x0C2

Phase 2 of CAN testing was to verify that the Parameter Messages (0x0C1 and 0x0C2) are used correctly to read and write parameters to the Rinehart controller. These parameters have many different functions. Some parameters are used to set non-volatile information (EEPROMdata). Other parameters are used to change functionality, and to monitor various operating parameters that are not part of the broadcast messages.

Message 0x0C1 is the ‘Read or Write Command’ sent by the MSC to the RMC. 0x0C2 is the confirmation response message from the RMC detailing whether or not the ‘Read or Write Command’ was successful.

To write a parameter data command in MotoTune, use message 0x0C1 and set the following:

- Set byte #2 to 1 (write) for ‘TxReadorWriteCMD’
- Enter in the address (0-99 General, 100-499 EEPROM) for ‘TxParameterAddress’
- Enter in the data content to be commanded (i.e. 0 or 1 for relay commands)

The RMC controller will then respond with message 0x0C2, and if successful, byte #2 will be set to 1. This can be read in MotoTune by including probe labeled ‘WriteResponse’ onto the PCM2 display.

Notice that message 0x0C1 is transmitted by PCM1 (located toward the right of Figure 4) with an address of 5, a write command and data content of 1. In response, notice that on the PCM1- display, the Rinehart Controller has received message 0x0C1 with the same information that was transmitted by PCM2(Figure 4).

The response to message 0x0C1 is the transmitted message 0x0C2 sent by the RMC to the MSC. The Rinehart Controller for these demonstration purposes is another MotoTron controller, and the message details were created and not real data. The purpose is to show that the supervisory controller (PCM2) receives message 0x0C2 successfully. This information is brought out on the PCM2 display which shows a 1 associated with probe labed ‘WriteResponse’. The one means that the RMC successfully accepted the 0x0C1 write command.

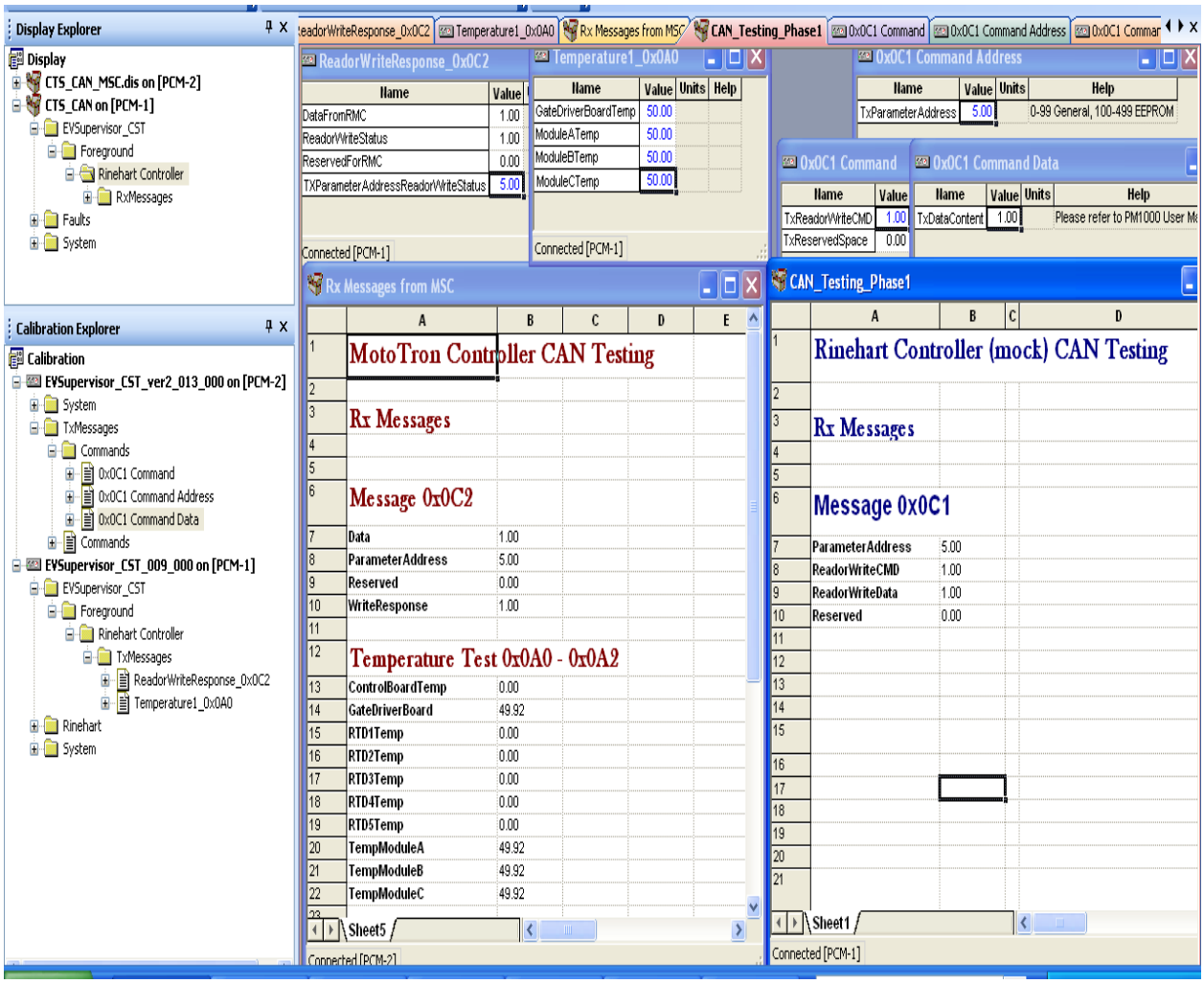

Figure 4: MotoTune Displays for PCM1 (Rinehart Controller) and PCM2 (MotoTron Controller).